

Jashore University of Science and Technology

Department of Physics

Bachelor of Science with Honours in Physics

2nd semester of 3rd year (2024–2025)

Course code: PHY 3205

Course title: Solid State Physics I

Assignment no.: 2

Date: 21 May 2026

Deadline for submission: 25 Jun 2026, 10:00 PM

1. Derive an expression for the attractive interaction between atoms in a solid composed of inert gases. Provide a detailed explanation of the underlying physical principles and assumptions.
2. What is the Madelung constant? Derive its expression for a one-dimensional ionic crystal and discuss its significance in determining the lattice energy of ionic solids.
3. Derive the dispersion relation for lattice vibrations in a one-dimensional crystal with a monatomic basis. Discuss the physical meaning of the acoustic phonon branch obtained from the relation.
4. Establish the Debye T^3 law for the heat capacity of solids at low temperatures. Define the Debye temperature and explain its significance in this model.
5. Define density of states. Derive an expression to calculate the density of states of a solid in 3D?
6. Describe the Fermi-Dirac distribution function, accompanied by an appropriate sketch of the distribution. How does it explain the occupancy of energy states in a fermion system? Why is the derivative of the Fermi-Dirac distribution function non-zero within the temperature range of $k_B T$ around the Fermi energy?